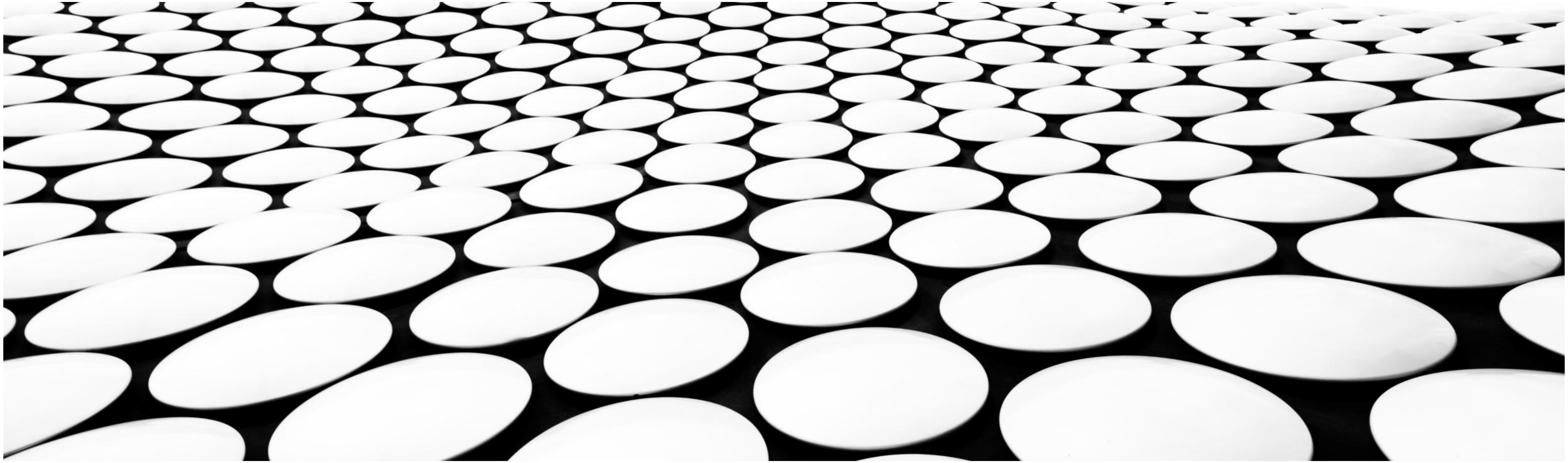
BIASES

Biases at every step of the Data Science Process



BIASES

- Introduction to biases
- Revisiting DS steps for biases
    - Data collection (primary/secondary)
    - Data cleaning and EDA
    - Modeling and evaluation
    - Reporting

DATA BIAS

THE BASICS

What is Data Bias?

The Basics is a series exploring the concepts and individuals essential to purposeful business.

Source

What is data bias?

Data bias occurs when data or information is limited in some way, painting an inaccurate representation of the population, or doesn't tell the full story. It has the capacity to impact individuals, businesses and even societies at large.

IBM

Authors

Julie Rogers
Writer

Alexandra Jonker
Editorial Content Lead

What is data bias?

Source

1. Cognitive bias
2. Automation bias
3. Confirmation bias
4. Exclusion bias
5. Historical bias
6. Implicit bias
7. Measurement bias
8. Reporting bias
9. Selection bias
10. Sampling bias

The 6 most common types of bias when working with data

The Metabase Team · Oct 08, 2021 in Analytics & BI · 9 min read

Source

1. Confirmation bias
2. Selection bias
3. Historical bias
4. Survivorship bias
5. Availability bias
6. Outlier bias

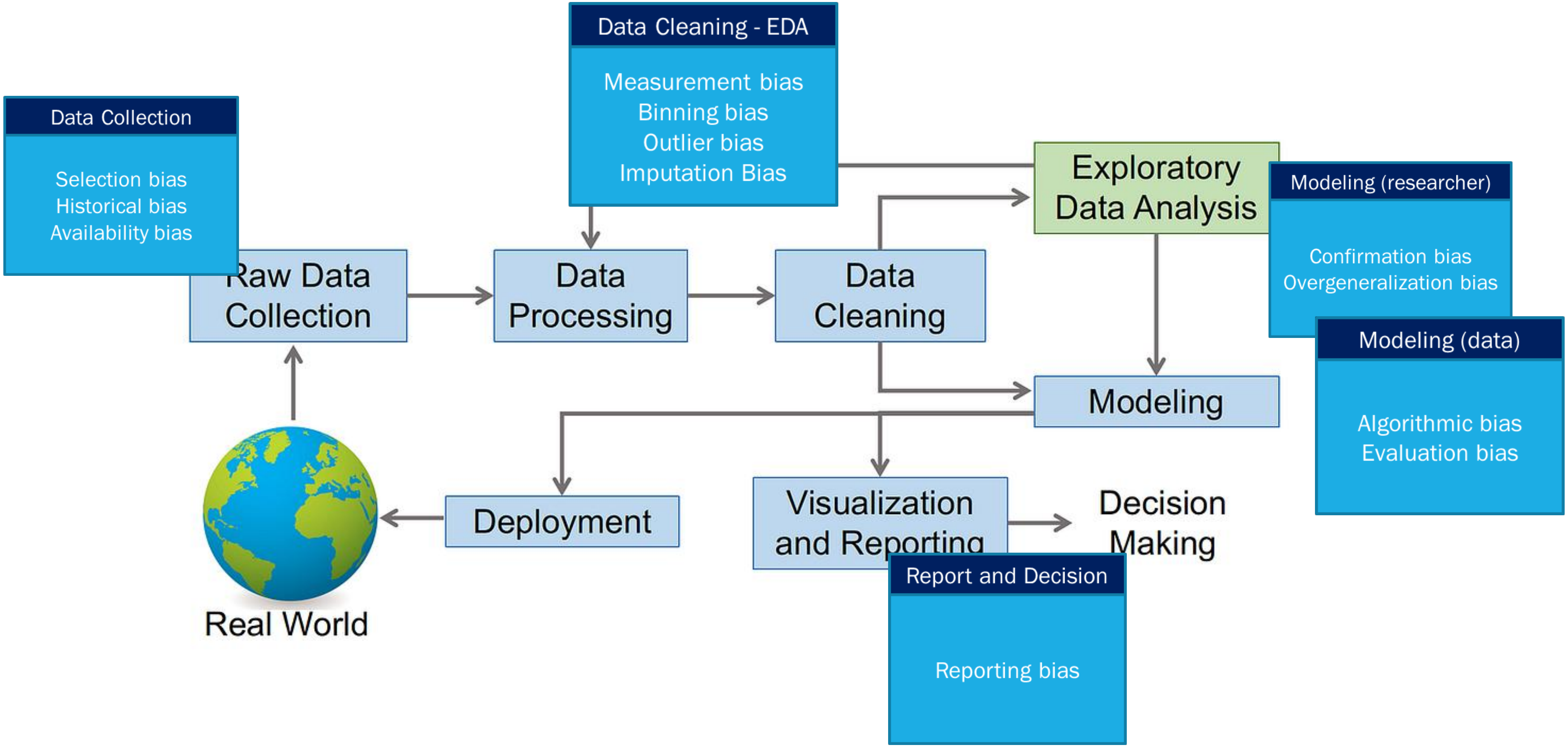
9 types of bias in data analysis and how to avoid them

Analytics can exhibit biases that affect the bottom line or incite social outrage through discrimination. It's important to address those biases before problems arise.

By George Lawton Published: 01 Jul 2024

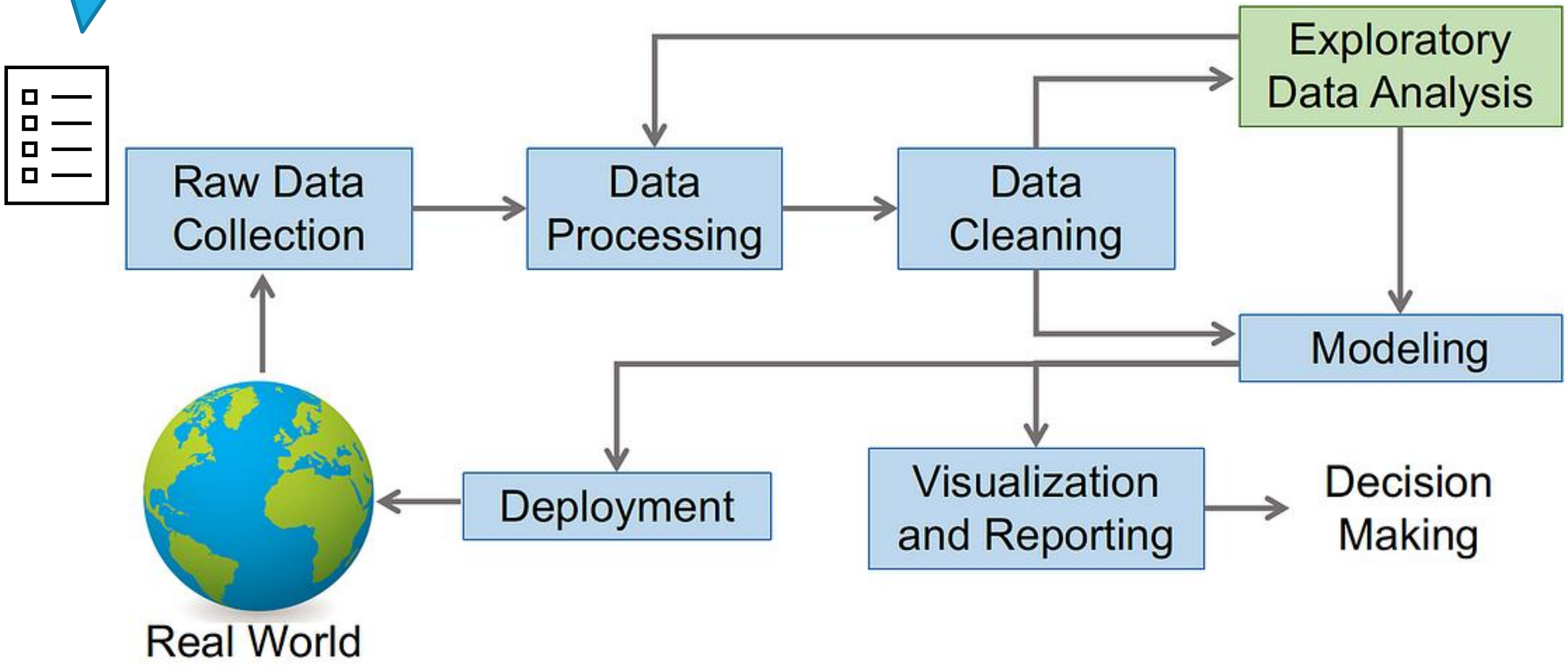
Source

1. Trained on the wrong thing
2. Confirmation bias
3. Availability bias
4. Temporal bias
5. AI infallibility bias
6. Optimist bias
7. Ghost in the machine bias
8. Preprocessing bias
9. Terminology bias


Data Cleaning - EDA
Measurement bias
Binning bias
Outlier bias
Imputation Bias
Data Collection
Selection bias
Historical bias
Availability bias
Raw Data Collection
Data Processing
Data Cleaning
Exploratory Data Analysis
Modeling (researcher)
Confirmation bias
Overgeneralization bias
Modeling (data)
Algorithmic bias
Evaluation bias
Modeling
Deployment
Visualization and Reporting
Decision Making
Report and Decision
Reporting bias
Real World

BIAS in Data Collection

Primary data collection



Selection Bias

Selection bias happens when the data set used for training is not representative enough, not large enough or too incomplete to sufficiently train a predictive system or obtain valid descriptive statistics.

Types of selection bias –

1. **Sampling bias:** occurs when randomization is not properly achieved during data collection.

2. **Convergence bias:** occurs when data is not selected in a representative manner.

3. **Participation bias:** occurs when the data is unrepresentative due to participations gaps in the data collection process.

Nice example

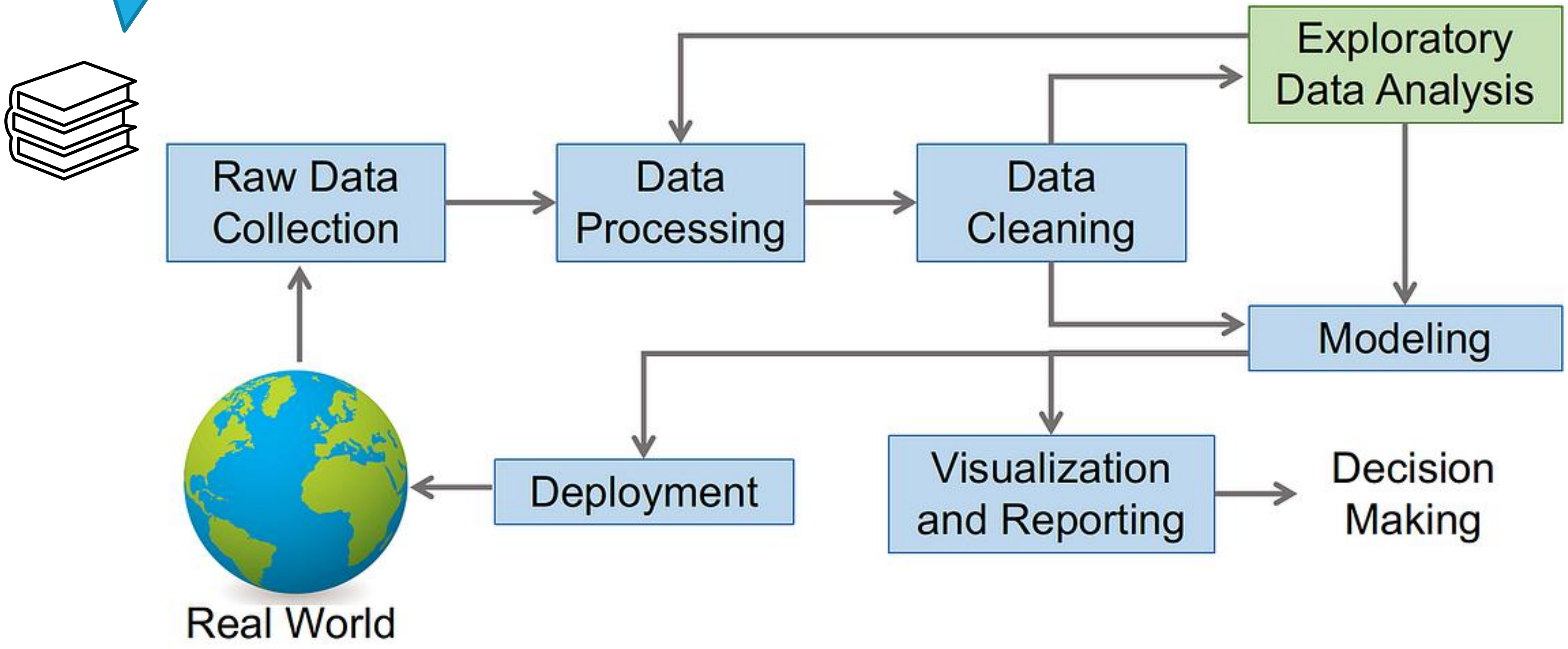
So let’s say Apple launched a new iPhone and on the same day Samsung launched a new Galaxy Note. You send out surveys to 1000 people to collect their reviews. Now instead of randomly selecting the responses for analysis, you decide to choose the first 100 customers that responded to your survey.

This will lead to **sampling bias** since those first 100 customers are more likely to be enthusiastic about the product and are likely to provide good reviews.

Next, if you decide to collect data by surveying only Apple customers by opting out of Samsung customers, you will induce a **convergence bias** in your dataset.

Lastly, you send the survey to 500 Apple and 500 Samsung customers. 400 Apple customers respond but only 100 Samsung customers respond. Now, this dataset would be underrepresenting the Samsung customers and would count towards **participation bias**.

Secondary data collection



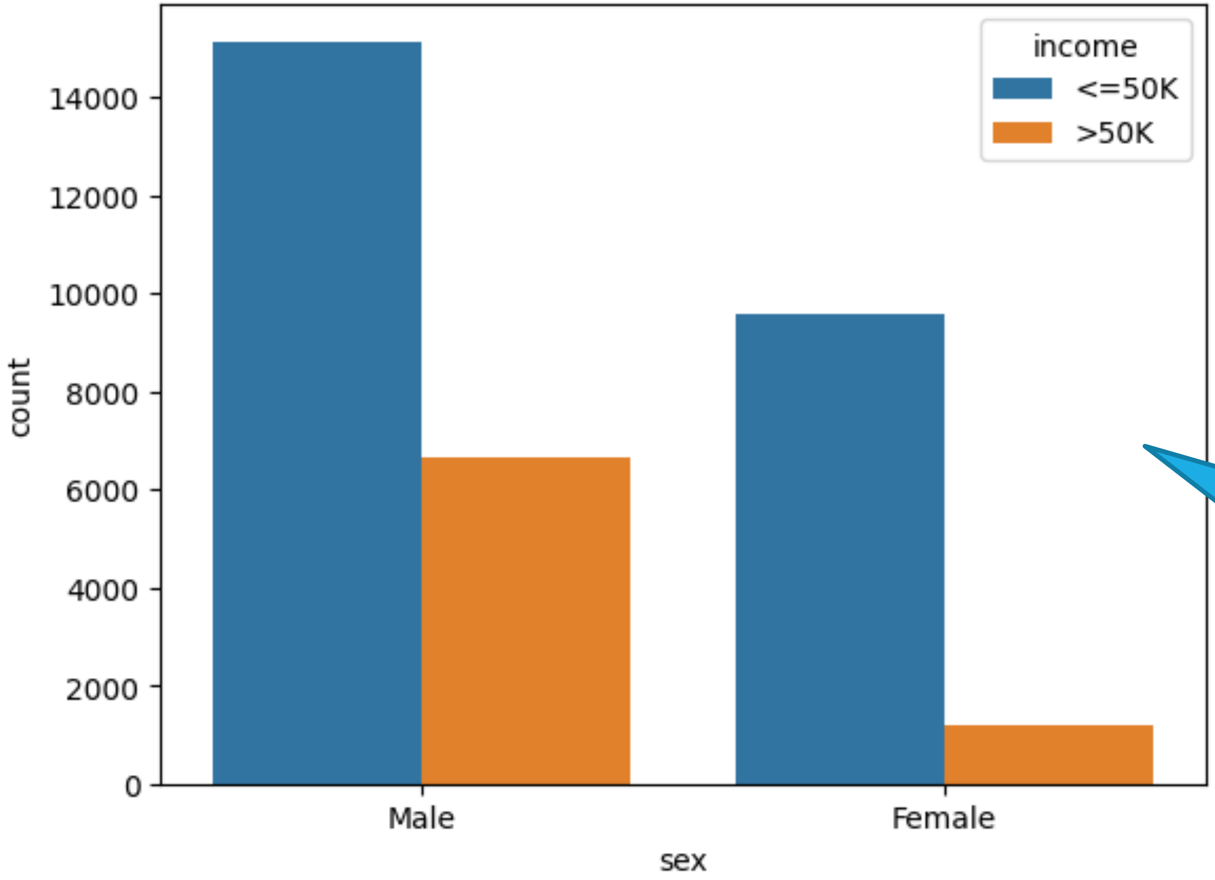
Historical Bias

Historical bias occurs when data reflects historical inequalities or biases that existed during data collection, as opposed to the current context. Historical data bias occurs when socio-cultural prejudices and beliefs are mirrored into systematic processes.

<u>Examples</u>
Manual (*human*) systems give certain groups of people poor credit ratings, and you're using that data to train the automatic system, the automatic system will replicate and may amplify the original system's biases.

Important to look into to prevent inequalities in prediction results

Census



Do we « throw away » the data, or qualify it within its historical context (1996)?

Availability Bias

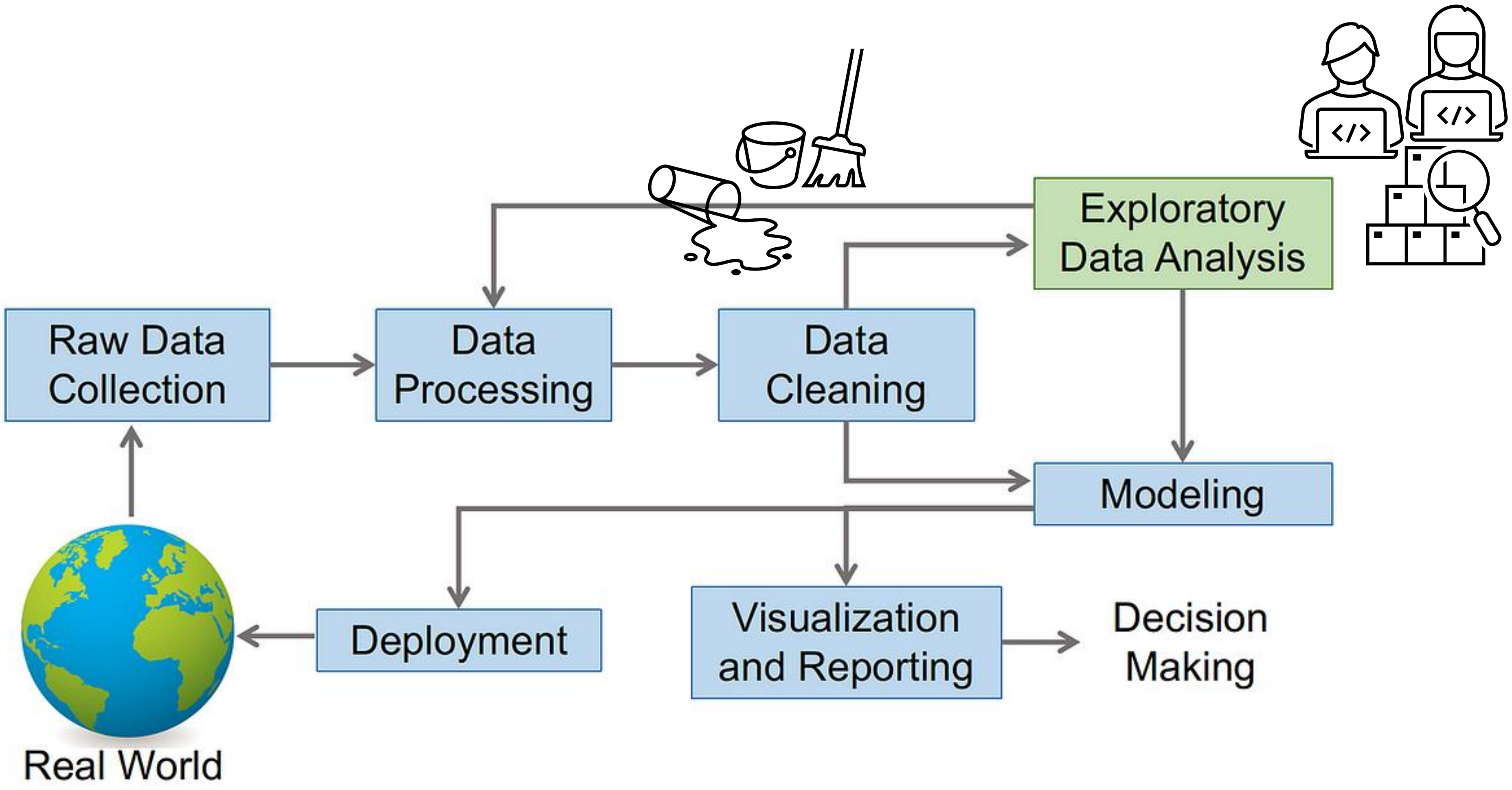
Or, even if a dataset is freely available, it might be « overused » because it is costly to create another one

Datasets previously in the public domain are being locked behind paywalls or are no longer available.

Depending upon the financial backing of the modelers and the types of data, future model results might be biased toward data sets that are still available for free within the public domain.

Issues about collecting data and then having « consumers » pay for it…

BIAS in Data Cleaning and EDA


Raw Data Collection
Data Processing
Data Cleaning
Exploratory Data Analysis
Modeling
Deployment
Visualization and Reporting
Decision Making
Real World

Measurement Bias

No contextualisation of measures

Measurement bias can occur when the accuracy or quality of the data differs across groups or when key study variables are inaccurately measured or classified.

Examples:

Using CGPAs to compare students without adjusting for the fact that some courses might be more difficult, and/or some universities might have more challenging programs.

Different calibration of instruments in different medical clinics (e.g. measuring blood pressure).

Binning Bias

Histograms group the values into intervals called bins, and the height of each bin in a histogram tells the number of points in that bin.

Bin size choice will impact the visualization and perhaps decision making.

For categorical data, we would talk of « group attribution bias »

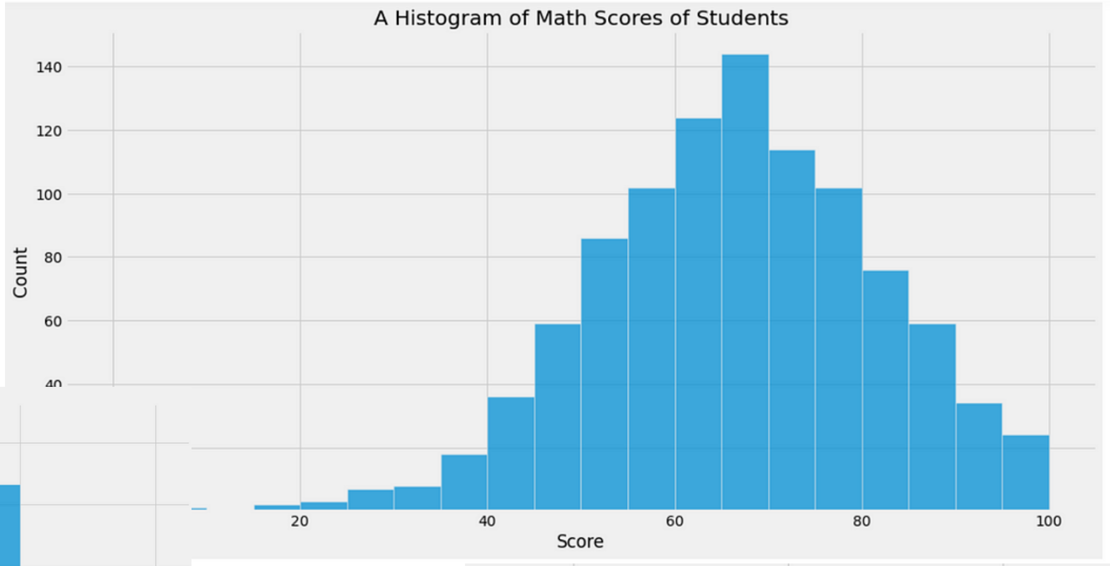

Member-only story

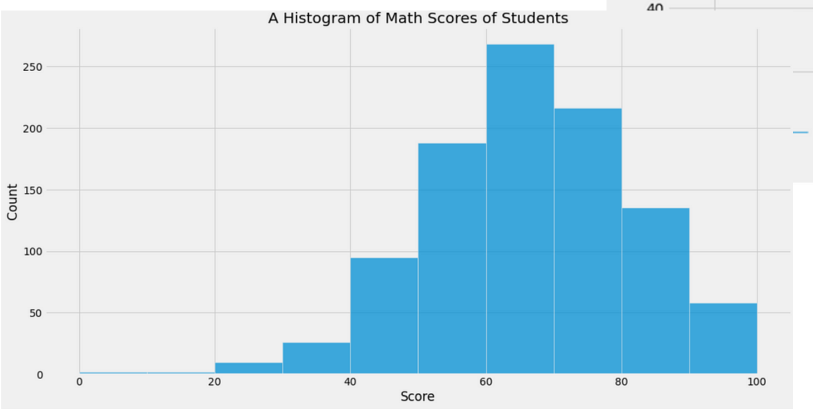
3 Best (Often Better) Alternatives To Histograms

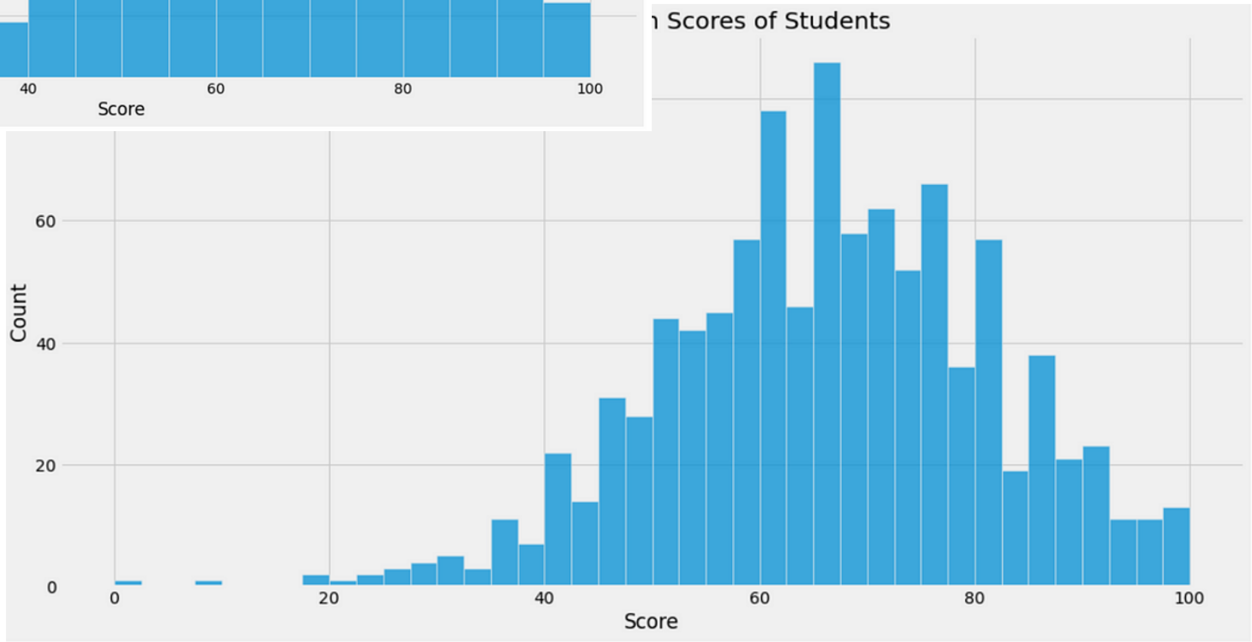
Avoid the most dangerous pitfall of histograms

Bex T. · Follow
Published in Towards AI · 10 min read · Sep 12, 2023

[Source]

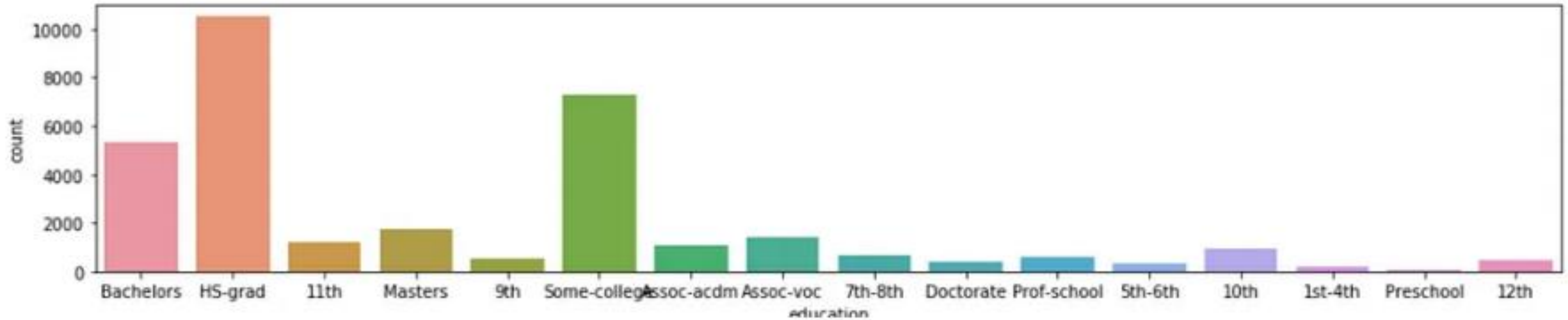






Census

Too many possible values, hard to interpret

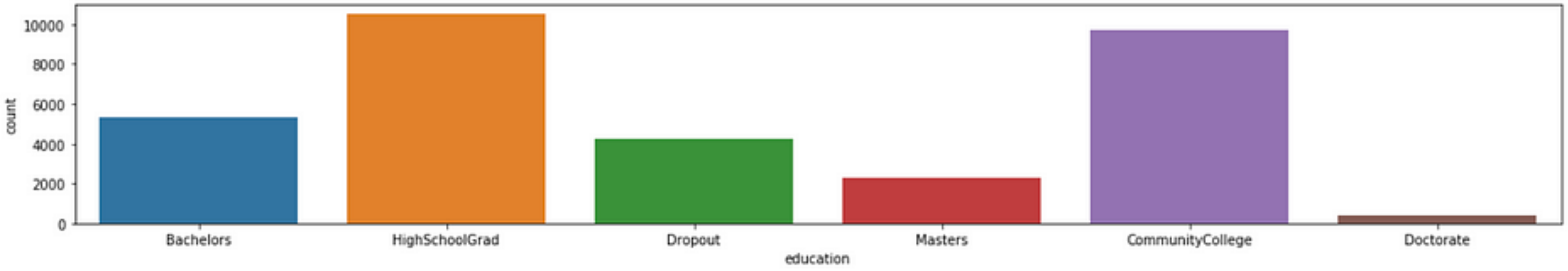


Census

In [28]:
```
#Grouping education
data['education'].replace('Preschool', 'Dropout',inplace=True)
data['education'].replace('10th', 'Dropout',inplace=True)
data['education'].replace('11th', 'Dropout',inplace=True)
data['education'].replace('12th', 'Dropout',inplace=True)
data['education'].replace('1st-4th', 'Dropout',inplace=True)
data['education'].replace('5th-6th', 'Dropout',inplace=True)
data['education'].replace('7th-8th', 'Dropout',inplace=True)
data['education'].replace('9th', 'Dropout',inplace=True)
data['education'].replace('HS-Grad', 'HighSchoolGrad',inplace=True)
data['education'].replace('HS-grad', 'HighSchoolGrad',inplace=True)
data['education'].replace('Some-college', 'CommunityCollege',inplace=True)
data['education'].replace('Assoc-acdm', 'CommunityCollege',inplace=True)
data['education'].replace('Assoc-voc', 'CommunityCollege',inplace=True)
data['education'].replace('Prof-school', 'Masters',inplace=True)

fig = plt.figure(figsize=(20,3))
sns.countplot(x="education", data=data)
```

Domain knowledge necessary to group categories. Can lead to « group attribution bias ».



Outlier Bias

Averages are a great place to hide uncomfortable truths. Some data is convenient to visualize as an average, but this simple operation hides the effect of outliers and anomalies, and skews our observations.

Examples:
Company with most people making between 50K and 100K,
but then a CEO making 2M.

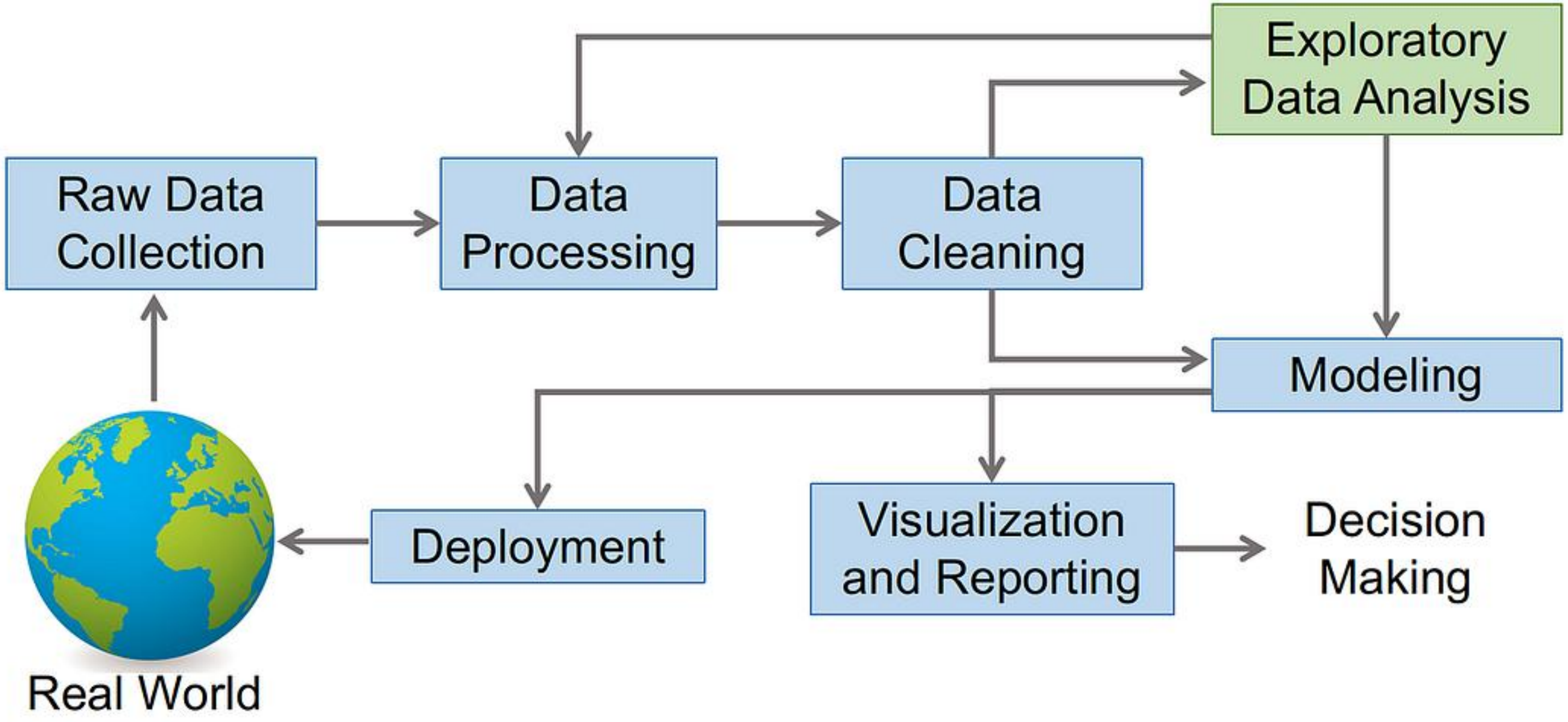
Imputation Bias

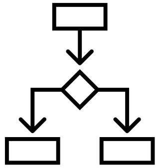
Replacing missing data is based on some underlying hypothesis (domain rules, distribution type, missing data type, etc). Any error in these hypotheses might lead to an inappropriate choice of imputation method.

Example of Mode Imputation

Customer	Age	Preferred Payment Method
A	30	Credit Card
B	45	PayPal
C	28	**Credit Card** (imputed)
D	50	Credit Card
E	35	Credit Card
F	40	**Credit Card** (imputed)

BIAS in Modeling and Evaluation





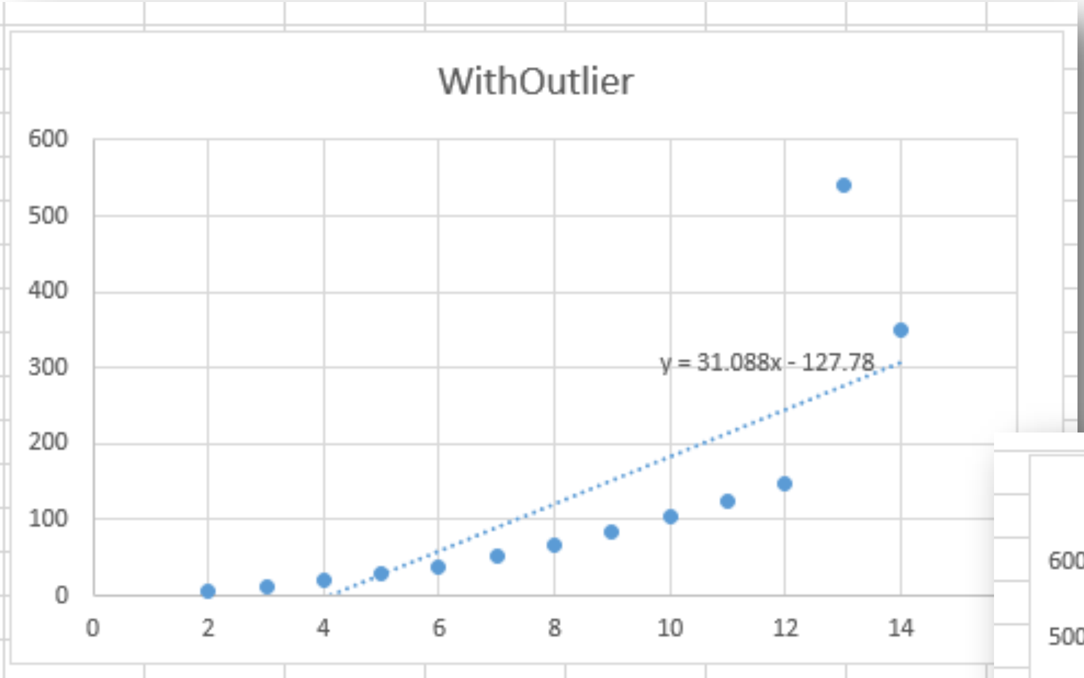
Choice of algorithm, segmentation, evaluation

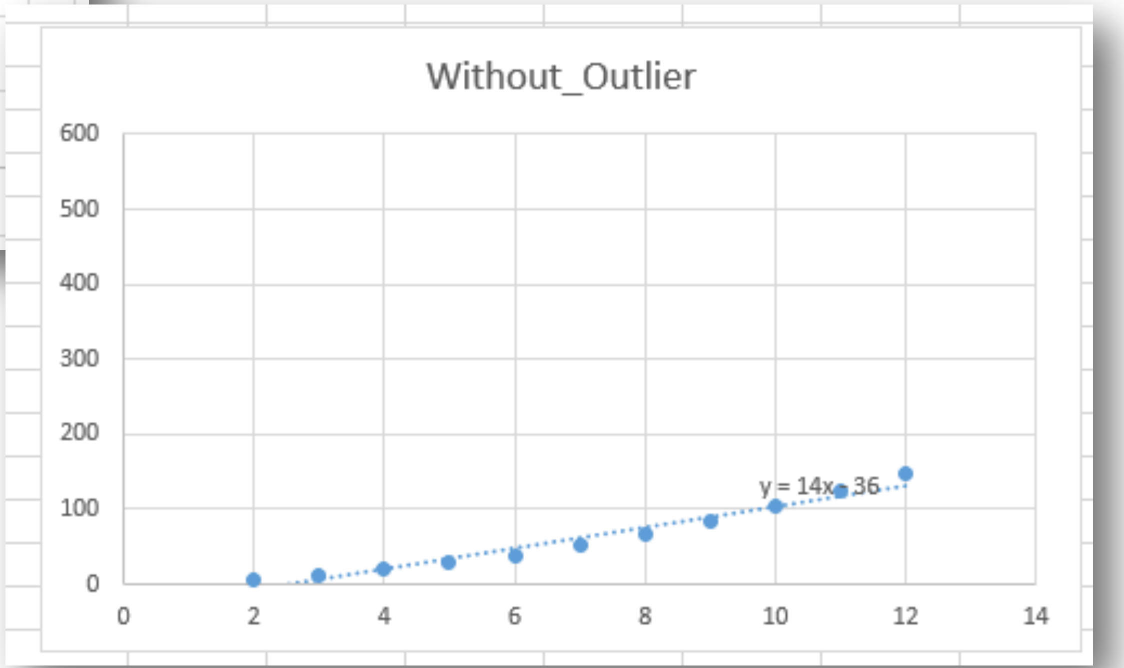
Algorithmic Bias

Understand the data first

A prediction algorithm should be chosen with an understanding of its pros/cons.

- Some predictive approaches, e.g. probabilistic approaches, are more sensitive to prior probability on classes.
- Some approaches are more sensitive to outliers than others. (e.g. regression models)
- Some approaches require a lot more data than others.



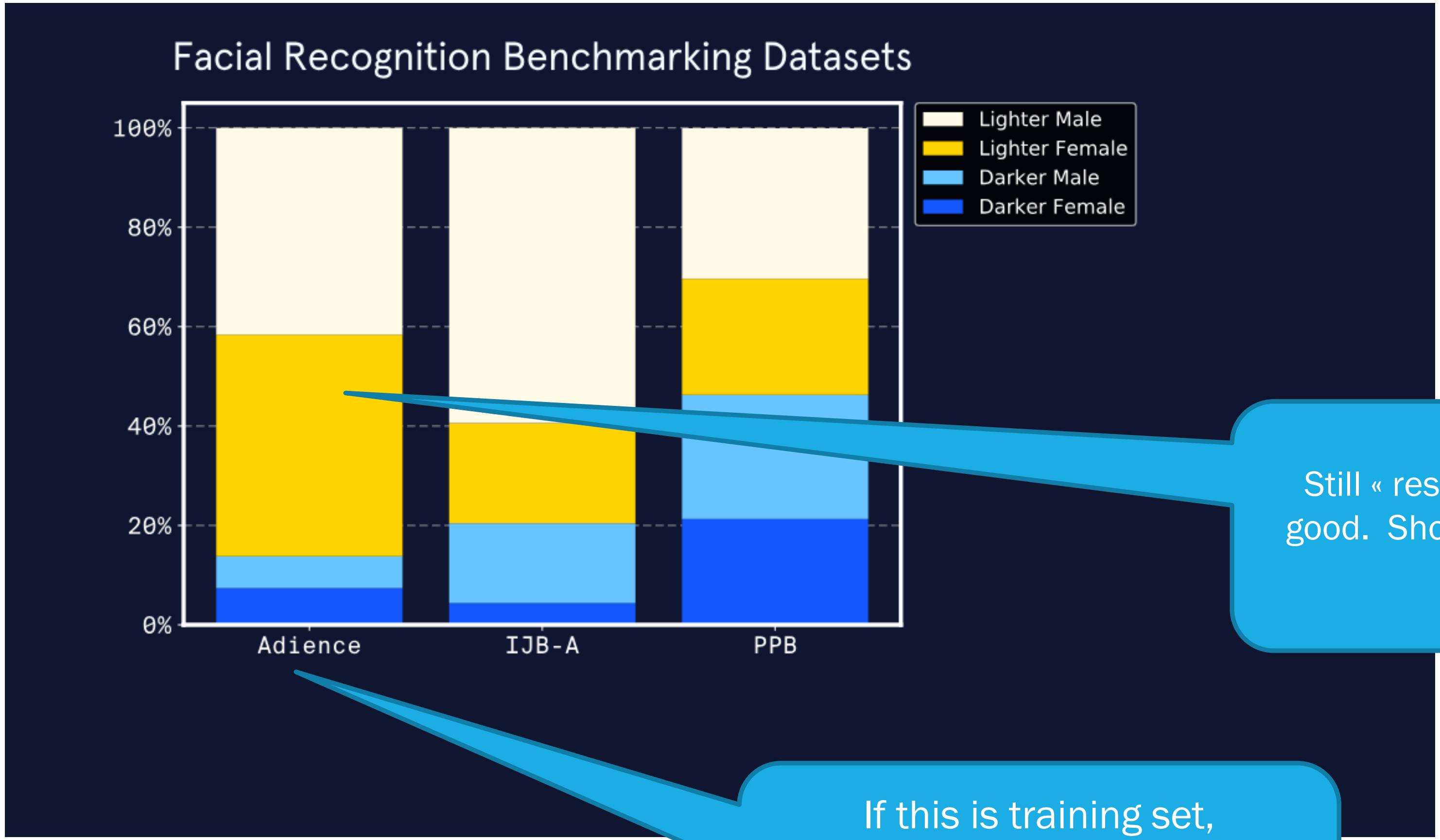


Regression models

Evaluation Bias

Testing an algorithm with a non-representative dataset leads to **evaluation bias**. Testing with a non-representative benchmarking dataset would give high overall accuracy scores, even if the algorithms were inaccurate for certain groups.

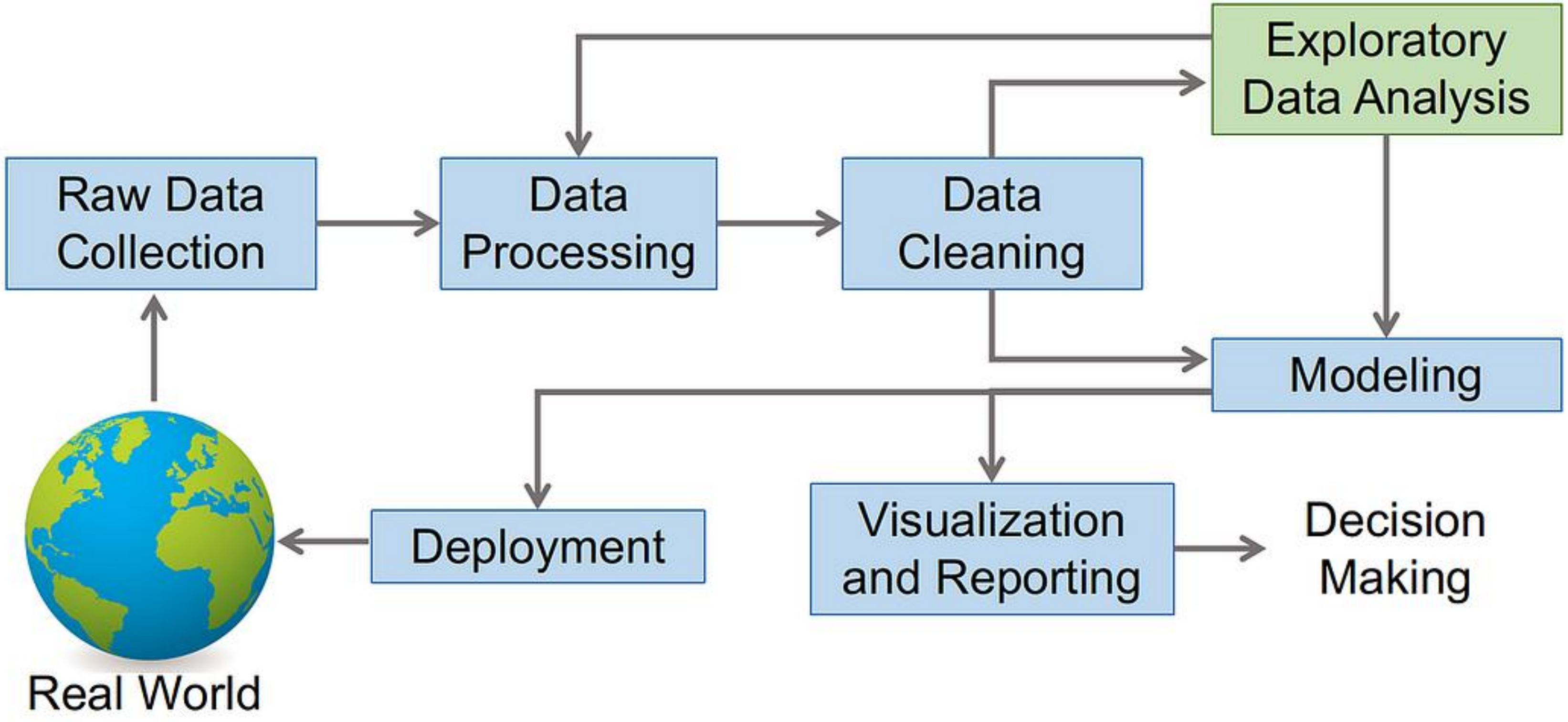
Pay attention to train/test...



Source

Still « results » overall will look good. Should publish results for each class.

If this is training set, discriminative features were not learned for the low-represented categories



Biases from the researcher

Confirmation bias

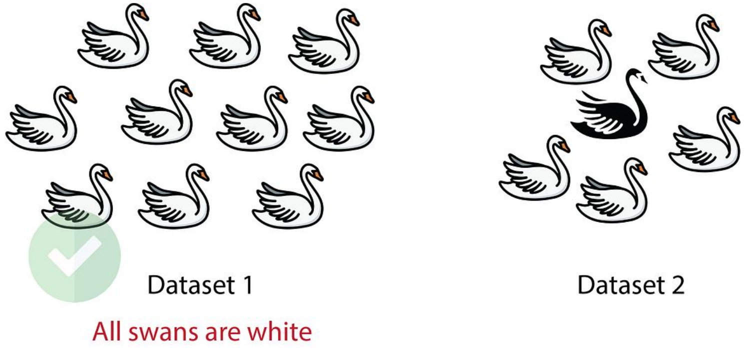
Also called experimenter's bias

A confirmation bias results when researchers choose only the data that supports their own hypothesis. Confirmation bias is found most often when evaluating results.

"If the results tend to confirm our hypotheses, we don't question them any further," said Theresa Kushner, partner at Business Data Leadership, a data consulting company.

"However, if the results don't confirm our hypotheses, we go out of our way to reevaluate the process, the data or the algorithms, thinking we must have made a mistake."

Overgeneralization bias

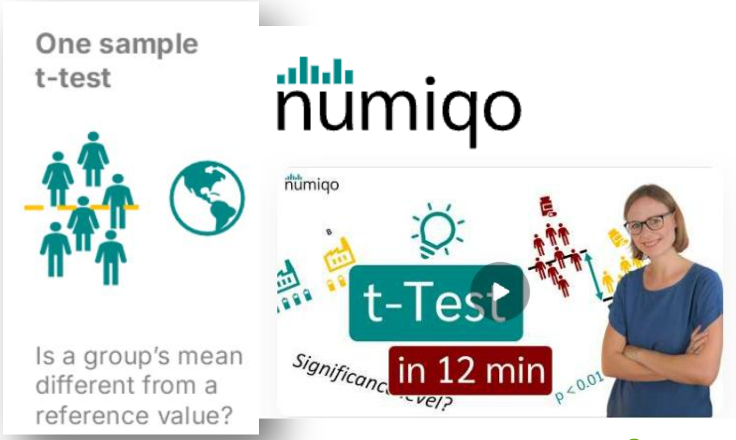


Overgeneralization occurs when you assume what you see in your dataset is :

- what you would see if you looked in any other dataset meant to assess the same information
- what you would see in the general population from which your dataset (sample) is taken

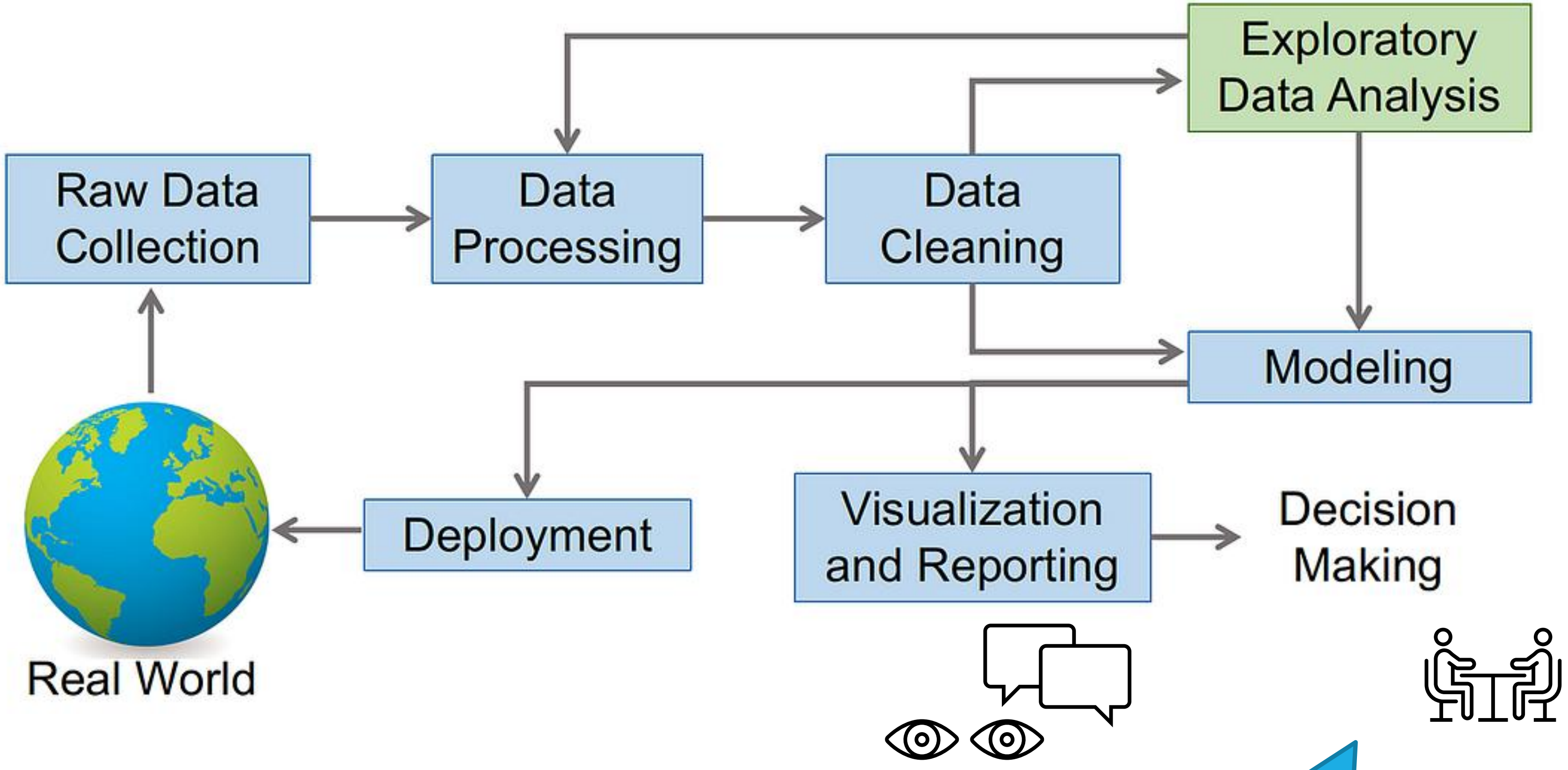
This one is sometimes called « transfer bias ».

Make sure results from sample applied to the full population



Source

BIAS in Reporting



Such biases influence
what is reported

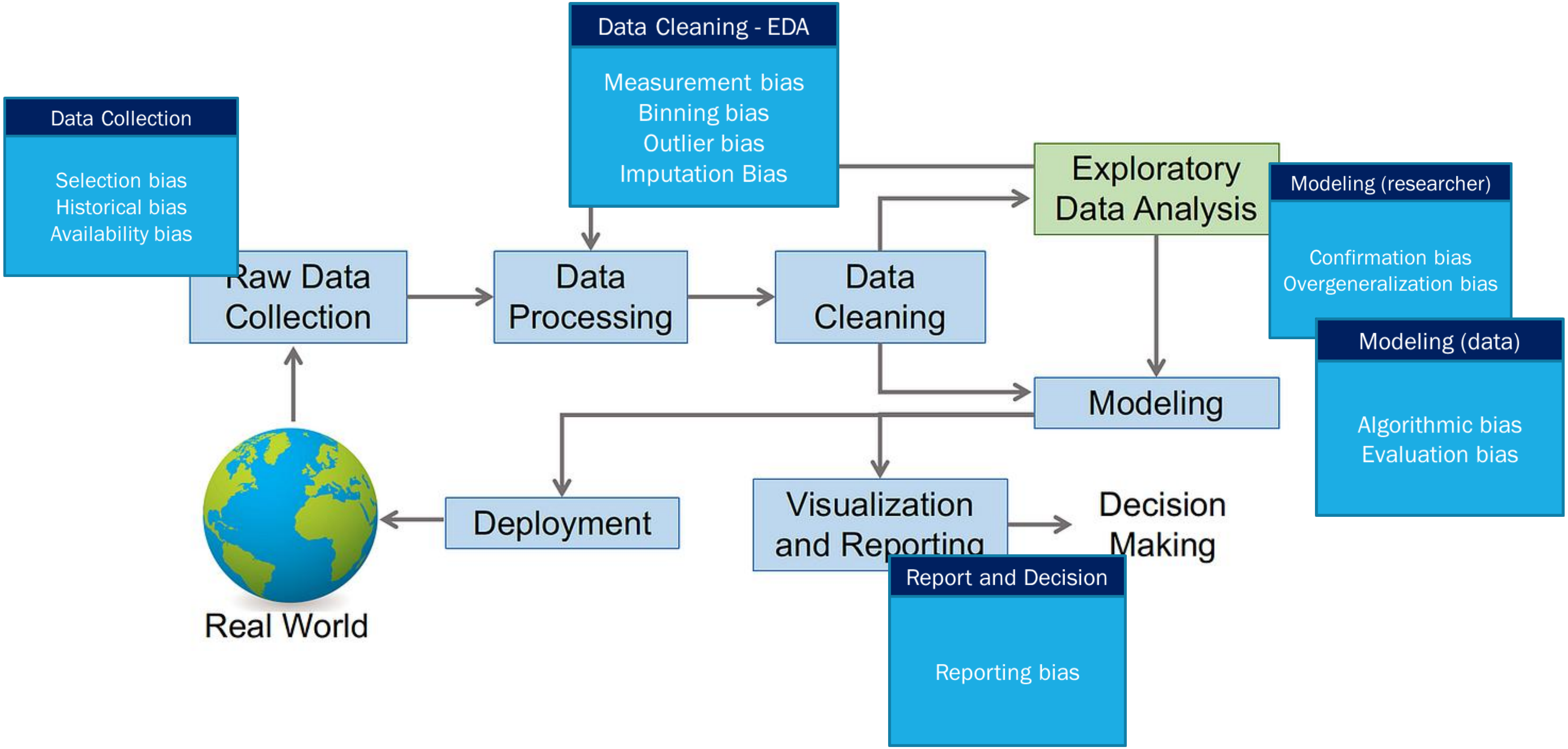
Reporting bias

Reporting bias (also known as selective reporting) takes place when only a **selection of results** or outcomes are reported. It is people's tendency to under-report all the information available.

This links back to confirmation bias too

Human tendency to only report or share results that affirm our beliefs or hypotheses, also known as "positive" results. Editors, publishers, and readers are also subject to reporting bias as positive results are published, read, and cited more often. To limit reporting bias, report negative results and cite others who do, too.

Sometimes called « survivorship bias ».


Data Cleaning - EDA
Measurement bias
Binning bias
Outlier bias
Imputation Bias
Data Collection
Selection bias
Historical bias
Availability bias
Raw Data Collection
Data Processing
Data Cleaning
Exploratory Data Analysis
Modeling (researcher)
Confirmation bias
Overgeneralization bias
Modeling (data)
Algorithmic bias
Evaluation bias
Modeling
Deployment
Visualization and Reporting
Decision Making
Report and Decision
Reporting bias
Real World

BIASES

- Introduction to biases
- Revisiting DS steps for biases
    - Data collection (primary/secondary)
    - Data cleaning and EDA
    - Modeling and evaluation
    - Reporting